

POTENTIALS

H_P: $\bar{U}$ is CONSERVATIVE

$$\bar{U} = \nabla f \leftarrow \text{SCALAR POTENTIAL}$$

f is DEFINED "UP TO A CONSTANT"

Assume that for some $\bar{U}$ there exist another $f^* = f + K, K \in \mathbb{R}^3$

$$\nabla f^* = \nabla (f + K) = \nabla f + \nabla K$$

$$\nabla f^* = \nabla f$$

GAUGE FREEDOM

H_P: $\bar{U}$ is SOLENOIDAL

$$\bar{U} = \nabla \times \bar{F} \leftarrow \text{VECTOR POTENTIAL}$$

$\bar{F}$ is defined "UP TO A CONSERVATIVE FIELD"

$$\bar{F}^* = \bar{F} + \nabla f \leftarrow \text{CONSERVATIVE FIELD}$$

$$\nabla \times \bar{F}^* = \nabla \times (\bar{F} + \nabla f) = \nabla \times \bar{F} + \nabla \times (\nabla f)$$

$$\nabla \times \bar{F}^* = \nabla \times \bar{F}$$

CURL of any CONSERVATIVE field is zero

COMPUTATION of CONSERVATIVE FIELDS

let $\bar{U}$ a vector field on D - $\bar{U}$ is conservative

$$\nabla \times \bar{U} = 0 \Rightarrow \bar{U} = \nabla f$$

$$\nabla \cdot \bar{U} = t \Rightarrow \nabla \cdot (\nabla f) = t \Rightarrow \boxed{\nabla^2 f = t}$$

SCALAR POISSON PROBLEM

LAPLACE OPERATOR

$$\nabla \cdot (\nabla (\dots)) \rightarrow \nabla^2 (\dots)$$

in CARTESIAN COORDINATES

$$\nabla^2 f =$$

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = t$$

ELLIPTIC PARTIAL differential Equation (PDE)

once f is known $\Rightarrow \bar{U} = \nabla f$

COMPUTATION of SOLENOIDAL FIELD

let $\bar{U}$ solenoidal field on D

$$\begin{cases} \nabla \cdot \bar{U} = 0 & \Rightarrow \bar{U} = \nabla \times \bar{F} \\ \nabla \times \bar{U} = \bar{T} & \Rightarrow \nabla \times (\nabla \times \bar{F}) = \bar{T} \end{cases}$$
$$\boxed{\nabla (\nabla \cdot \bar{F}) - \nabla^2 \bar{F} = \bar{T}}$$

RECALL: any vector potential is defined up to a conservative field

$$\nabla (\nabla \cdot \bar{F}) \Rightarrow \text{CONSERVATIVE FIELD} \Rightarrow \text{no influence on } \nabla \times \bar{F}$$

GAUGE FREEDOM $\Rightarrow$ SET $\nabla \cdot \bar{F} = 0$ without changing $\bar{U}$

$$\boxed{\nabla^2 \bar{F} = -\bar{T}}$$

VECTOR
POISSON
PROBLEM

$$\begin{cases} \nabla^2 F_x = -T_x \\ \nabla^2 F_y = -T_y \\ \nabla^2 F_z = -T_z \end{cases}$$

3 scalar
poisson problems

once $\bar{F}$ is known $\rightarrow \bar{U} = \nabla \times \bar{F}$

Helmholtz theorem

any vector field can be decomposed into a CONSERVATIVE component and a SOLENOIDAL component

$$\bar{U} = \bar{U}_c + \bar{U}_s$$

$\uparrow$ CONSERVATIVE $\uparrow$ SOLENOIDAL

$$\begin{array}{ll} \bar{U} = \nabla f & \bar{U} = \nabla \times \bar{F} \\ \nabla^2 f = t & \nabla^2 \bar{F} = -\bar{T} \end{array}$$

$$u = \nabla f$$

$$\nabla^2 f = t$$

$$u = \nabla \phi$$

$$\nabla^2 \phi = -T$$

RECAP of EM

Lorentz Force

q be a point charge

$$\vec{F} = q (\vec{E} + \vec{v} \times \vec{B})$$

force of EM origin

vector product

velocity of point charge

if $\vec{v} \times \vec{B} = 0 \Rightarrow \vec{F} = q\vec{E} \quad [N = \frac{J}{m}]$

$$\vec{E} = \vec{F}/q = [N/C] = [\frac{J}{C \cdot m}] = [V/m]$$

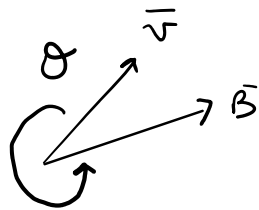
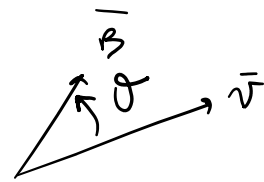
Forces per unit of charge

[V]

if $\vec{E} = 0 \Rightarrow \vec{F} = q \vec{v} \times \vec{B}$

$$|\vec{F}| \downarrow$$

$$F = q v B \sin \theta$$



$$B = \frac{F}{q v \sin(\theta)} = [N / (C \cdot m/s)] = [T]$$

forces per unit of charge and velocity

Direction $\Rightarrow$ right-hand rule

Example

